

# Fast algorithms for hierarchical matrices

Introductory course on numerical methods and fast algorithms

Sep 22, 2025

# What is fast?

Suppose  $A \in \mathbb{C}^{N \times N}$ , and,  $v \in \mathbb{C}^N$

- **Matvec**  $A \cdot v : O(N^2)$
- **Inversion**  $A^{-1} : O(N^3)$
- **Determinants**  $\det A : O(N^3)$

For a given task, an algorithm is fast if it's runtime beats the asymptotic complexity

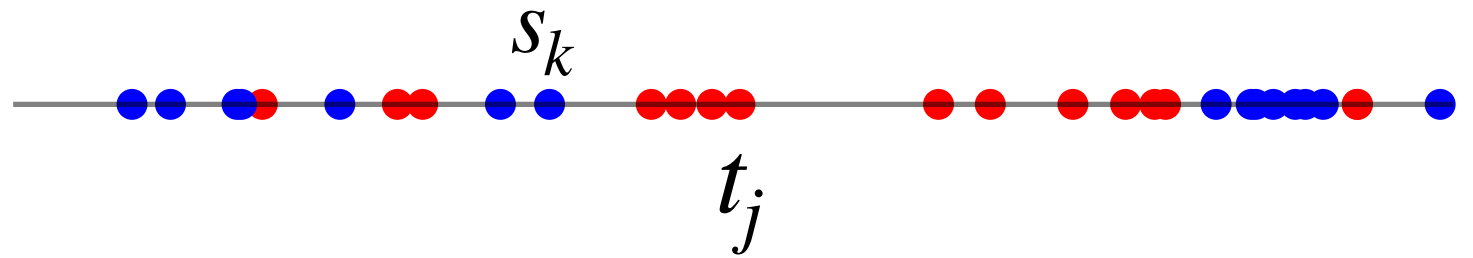
**The dream:**  $O(N \log^s N)$

## Examples

- Sparse matrices, matvecs in  $O(kN)$ , if well-conditioned, inverse in  $O(kN)$
- FFT matrices, matvecs in  $O(N \log N)$ , inverse analytically known, and inverse application in  $O(N \log N)$
- FMM matrices, matvecs in  $O(N)$

# Dense matrices $\neq$ Data dense

$$\begin{aligned} A_{j,k} &= \delta_{j,k} + \cos(t_j - s_k) \\ &= \delta_{j,k} + \cos(t_j)\cos(s_k) + \sin(t_j)\sin(s_k) \end{aligned}$$



- **Matvec**  $b = A \cdot v$  :  $O(N^2)$

**Step 1:**

$$W_1 = \sum_{k=1}^N \cos(s_k) v_k, \quad W_2 = \sum_{k=1}^N \sin(s_k) v_k$$

**Step 2:**

$$b_j = v_j + \cos(t_j) W_1 + \sin(t_j) W_2 \quad O(N)!$$

$$A = I + UV^T, \quad U = \begin{bmatrix} \cos(t_1) & \sin(t_1) \\ \cos(t_2) & \sin(t_2) \\ \vdots & \vdots \\ \cos(t_N) & \sin(t_N) \end{bmatrix}, \quad V = \begin{bmatrix} \cos(s_1) & \sin(s_1) \\ \cos(s_2) & \sin(s_2) \\ \vdots & \vdots \\ \cos(s_N) & \sin(s_N) \end{bmatrix}$$

- **Inversion**  $A^{-1}$  :  $O(N^3)$

**Sherman Morrison Woodbury formula:**  $A^{-1} = I - U(I_2 + V^T U)^{-1} V^T$   $O(N)!$

- **Determinants**  $\det A$  :  $O(N^3)$

**Sylvester formula formula:**  $\det A = \det(I_2 + V^T U)$   $O(N)!$

# Two-point boundary value problems

Given  $p(x), q(x), f(x), \alpha, \beta$ , find  $u(x)$  which satisfies

$$u''(x) + p(x)u'(x) + q(x)u(x) = f(x) \quad 0 < x < 1.$$
$$u(0) = \alpha, \quad u(1) = \beta.$$

$u \rightarrow u(x) - \alpha - (\beta - \alpha)x$  satisfies

$$u''(x) + p(x)u'(x) + q(x)u(x) = \tilde{f}(x) \quad 0 < x < 1.$$
$$u(0) = 0, \quad u(1) = 0.$$

$G(x, y)$  : Green's function for

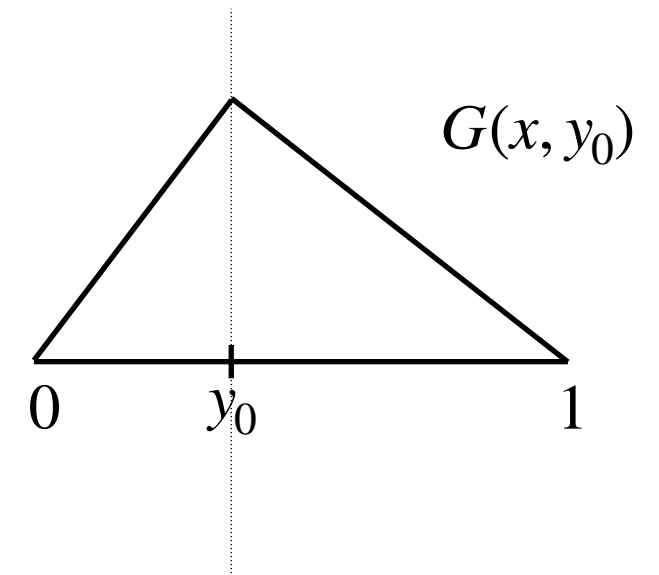
$$u''(x) = \delta_y \quad 0 < x < 1.$$
$$u(0) = 0, \quad u(1) = 0.$$

$$G(x, y) = \begin{cases} x(1 - y) & 0 \leq x < y \leq 1 \\ y(1 - x) & 0 \leq y < x \leq 1 \end{cases}$$

**Integral formulation:**  $u(x) = \int_0^1 G(x, y)\sigma(y) dy$ ,  $\sigma$  unknown density

- Boundary conditions ✓
- ODE yields following integral equation

$$\sigma(x) + p(x) \int_0^1 \frac{\partial G}{\partial x}(x, y)\sigma(y) dy + q(x) \int_0^1 G(x, y)\sigma(y) dy = \tilde{f}, \quad 0 < x < 1$$



# Structure of off-diagonal blocks

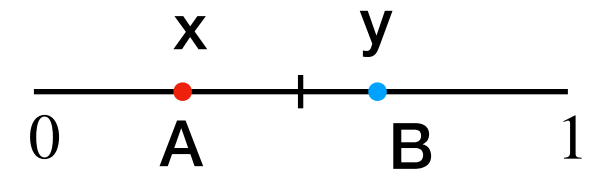
$$\sigma(x) + p(x) \int_0^1 \frac{\partial G}{\partial x}(x, y) \sigma(y) dy + q(x) \int_0^1 G(x, y) \sigma(y) dy = P\sigma = \tilde{f}$$

$$\begin{bmatrix} P_{AA} & P_{AB} \\ P_{BA} & P_{BB} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \end{bmatrix}$$

$$P_{AB}\sigma_B = p(x) \int_B (1 - y) \sigma(y) dy + q(x) \cdot x \int_B (1 - y) \sigma(y) dy$$

$$P_{AB} = (p(x) + q(x) \cdot x) \int_B (1 - y) \cdot * dy$$

$$u_A \quad v_B^T$$



$$G(x, y) = \begin{cases} x(1 - y) & 0 \leq x < y \leq 1 \\ y(1 - x) & 0 \leq y < x < \leq 1 \end{cases}$$

Rank 1

# Structure of off-diagonal blocks

$$\sigma(x) + p(x) \int_0^1 \frac{\partial G}{\partial x}(x, y) \sigma(y) dy + q(x) \int_0^1 G(x, y) \sigma(y) dy = P\sigma = \tilde{f}$$

$$\begin{bmatrix} P_{AA} & P_{AB} \\ P_{BA} & P_{BB} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \end{bmatrix}$$

$$P_{AB}\sigma_B = p(x) \int_B (1 - y) \sigma(y) dy + q(x) \cdot x \int_B (1 - y) \sigma(y) dy$$

$$P_{AB} = (p(x) + q(x) \cdot x) \int_B (1 - y) \cdot * dy$$

$u_A$

$v_B^T$

Rank 1

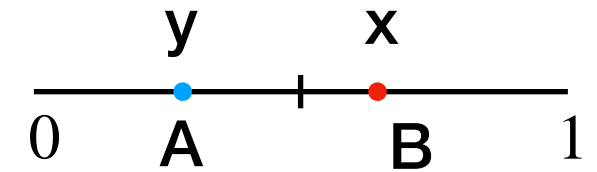
$$P_{BA}\sigma_A = -p(x) \int_A y \sigma(y) dy + q(x) \cdot (1 - x) \int_A y \sigma(y) dy$$

$$P_{AB} = (-p(x) + q(x) \cdot (1 - x)) \int_A y \cdot * dy$$

$u_B$

$v_A^T$

Rank 1



$$G(x, y) = \begin{cases} x(1 - y) & 0 \leq x < y \leq 1 \\ y(1 - x) & 0 \leq y < x < \leq 1 \end{cases}$$

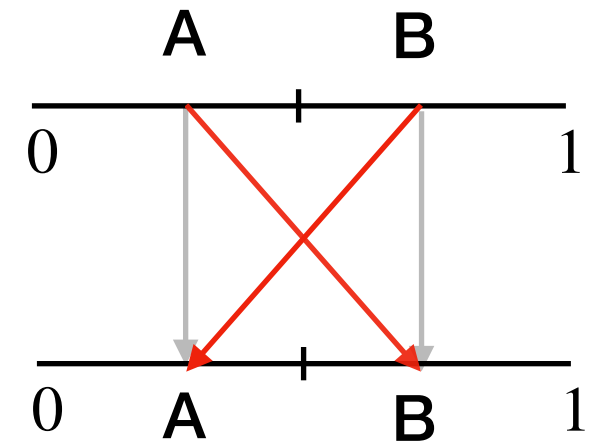
## Faster? Matvec and inverse

$$\begin{bmatrix} P_{AA} & u_A v_B^T \\ u_B v_A^T & P_{BB} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \end{bmatrix}$$

### Matvec

#### Step 1:

Compute  $u_B v_A^T \sigma_A, u_A v_B^T \sigma_B$   $O(N)$



# Faster? Matvec and inverse

$$\begin{bmatrix} P_{AA} & u_A v_B^T \\ u_B v_A^T & P_{BB} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \end{bmatrix}$$

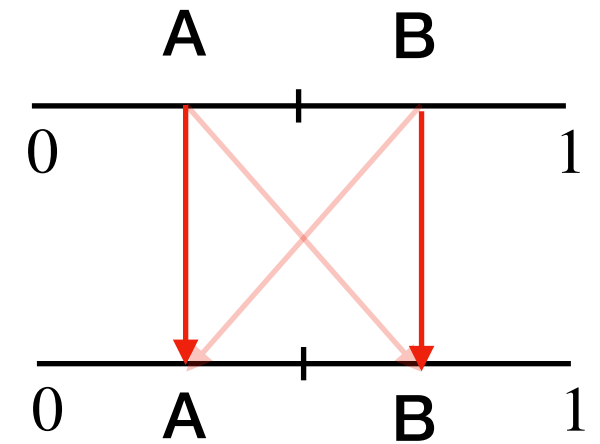
## Matvec

### Step 1:

Compute  $u_B v_A^T \sigma_A, u_A v_B^T \sigma_B$   $O(N)$

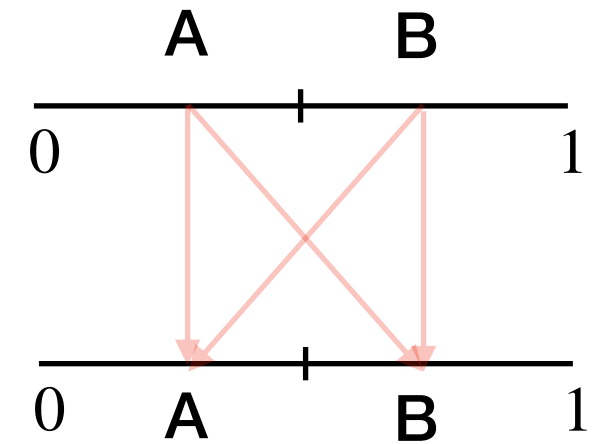
### Step 2:

Compute  $P_{AA} \sigma_A, P_{BB} \sigma_B$   $O(N^2/2)$



# Faster? Matvec and inverse

$$\begin{bmatrix} P_{AA} & u_A v_B^T \\ u_B v_A^T & P_{BB} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \end{bmatrix}$$



## Matvec

### Step 1:

Compute  $u_B v_A^T \sigma_A, u_A v_B^T \sigma_B$   $O(N)$

### Step 2:

Compute  $P_{AA} \sigma_A, P_{BB} \sigma_B$   $O(N^2/2)$

## Inverse

$D$

$UV^T$

$$\begin{bmatrix} P_{AA} & u_A v_B^T \\ u_B v_A^T & P_{BB} \end{bmatrix} = \begin{bmatrix} P_{AA} & 0 \\ 0 & P_{BB} \end{bmatrix} + \begin{bmatrix} u_A & 0 \\ 0 & u_B \end{bmatrix} \begin{bmatrix} v_A^T & 0 \\ 0 & v_B^T \end{bmatrix}$$

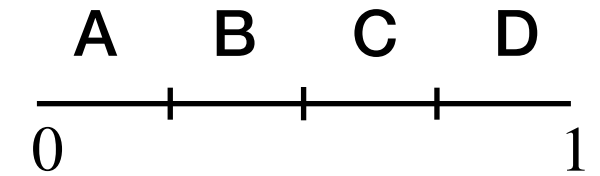
$N \times 2 \quad 2 \times N$

$$P^{-1} = \begin{bmatrix} P_{AA}^{-1} & 0 \\ 0 & P_{BB}^{-1} \end{bmatrix} - \begin{bmatrix} P_{AA}^{-1} u_A & 0 \\ 0 & P_{BB}^{-1} u_B \end{bmatrix} \left( I_2 + \begin{bmatrix} v_A^T P_{AA}^{-1} u_A & 0 \\ 0 & v_B^T P_{BB}^{-1} u_B \end{bmatrix} \right)^{-1} \begin{bmatrix} v_A^T P_{AA}^{-1} & 0 \\ 0 & v_B^T P_{BB}^{-1} \end{bmatrix} \quad O(N^3/4)$$

$D^{-1} - D^{-1} U (I + V^T D^{-1} U)^{-1} V^T D^{-1}$

# Structure of off-diagonal blocks

$$\begin{bmatrix} P_{AA} & P_{AB} & P_{AC} & P_{AD} \\ P_{BA} & P_{BB} & P_{BC} & P_{BD} \\ P_{CA} & P_{CB} & P_{CC} & P_{CD} \\ P_{DA} & P_{DB} & P_{DC} & P_{DD} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \\ \sigma_C \\ \sigma_D \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \\ \tilde{f}_D \end{bmatrix}$$



All off-diagonal blocks are rank 1

$$\begin{bmatrix} P_{AA} & u_{A,R}v_{B,L}^T & u_{A,R}v_{C,L}^T & u_{A,R}v_{D,L}^T \\ u_{B,L}v_{A,R}^T & P_{BB} & u_{B,R}v_{C,L}^T & u_{B,R}v_{D,L}^T \\ u_{C,L}v_{A,R}^T & u_{C,L}v_{B,R}^T & P_{CC} & u_{C,R}v_{D,L}^T \\ u_{D,L}v_{A,R}^T & u_{D,L}v_{B,R}^T & u_{D,L}v_{C,R}^T & P_{DD} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \\ \sigma_C \\ \sigma_D \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \\ \tilde{f}_D \end{bmatrix}$$

$$u_{I,L} = -p(x) + (1-x) \cdot q(x), \quad x \in I$$

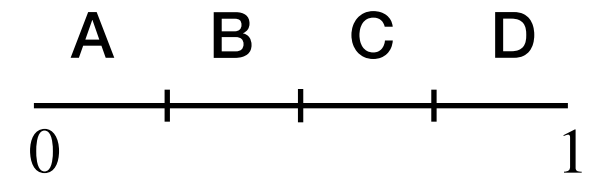
$$v_{I,L} = 1 - y, \quad y \in I$$

$$u_{I,R} = p(x) + x \cdot q(x), \quad x \in I$$

$$v_{I,R} = y, \quad y \in I$$

# Block separable form

$$\begin{bmatrix} P_{AA} & \mathbf{u}_{A,R} \mathbf{v}_{B,L}^T & u_{A,R} v_{C,L}^T & u_{A,R} v_{D,L}^T \\ u_{B,L} v_{A,R}^T & P_{BB} & u_{B,R} v_{C,L}^T & u_{B,R} v_{D,L}^T \\ u_{C,L} v_{A,R}^T & u_{C,L} v_{B,R}^T & P_{CC} & u_{C,R} v_{D,L}^T \\ u_{D,L} v_{A,R}^T & u_{D,L} v_{B,R}^T & u_{D,L} v_{C,R}^T & P_{DD} \end{bmatrix}$$



$$\begin{bmatrix} P_{AA} & \mathbf{u}_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \mathbf{v}_B^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ \mathbf{u}_B \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & P_{BB} & \mathbf{u}_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & \mathbf{u}_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & P_{CC} & u_C \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_C^T & P_{DD} \end{bmatrix}$$

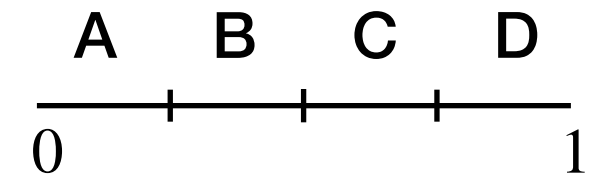
$$\mathbf{u}_I = (u_{I,L} \quad u_{I,R})$$

$$\mathbf{v}_I^T = \begin{pmatrix} v_{I,L}^T \\ v_{I,R}^T \end{pmatrix}$$

$$P_{i,j} = u_i S_{i,j} v_j^T$$

# Block separable form

$$\begin{bmatrix} P_{AA} & u_{A,R} v_{B,L}^T & u_{A,R} v_{C,L}^T & u_{A,R} v_{D,L}^T \\ u_{B,L} v_{A,R}^T & P_{BB} & u_{B,R} v_{C,L}^T & u_{B,R} v_{D,L}^T \\ u_{C,L} v_{A,R}^T & u_{C,L} v_{B,R}^T & P_{CC} & u_{C,R} v_{D,L}^T \\ u_{D,L} v_{A,R}^T & u_{D,L} v_{B,R}^T & u_{D,L} v_{C,R}^T & P_{DD} \end{bmatrix}$$



$$\begin{bmatrix} P_{AA} & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_B^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_B \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & P_{BB} & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & P_{CC} & u_C \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_C^T & P_{DD} \end{bmatrix}$$

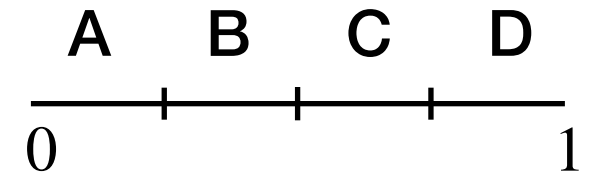
$$u_I = (u_{I,L} \quad u_{I,R})$$

$$v_I^T = \begin{pmatrix} v_{I,L}^T \\ v_{I,R}^T \end{pmatrix}$$

$$P_{i,j} = u_i S_{i,j} v_j^T$$

## Block separable form

$$\begin{bmatrix} P_{AA} & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_B^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_B \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & P_{BB} & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & P_{CC} & u_C \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_C^T & P_{DD} \end{bmatrix}$$



$$P_{i,j} = u_i S_{i,j} v_j^T$$

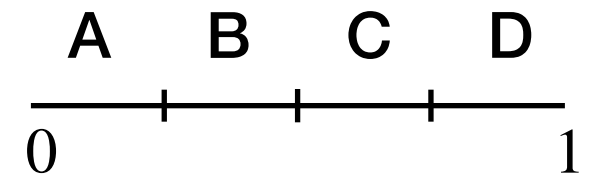
$$P = U S V^T + D$$

The diagram illustrates the block separable form  $P = U S V^T + D$  using 4x4 matrices. 
 

- $P$  is a solid black 4x4 matrix.
- $=$  is the equality symbol.
- $U$  is a 4x4 matrix with a single column of four black squares on the diagonal.
- $S$  is a 4x4 matrix with a checkerboard pattern of black and white squares.
- $V^T$  is a 4x4 matrix with a single row of four black squares on the diagonal.
- $+$  is the addition symbol.
- $D$  is a 4x4 matrix with a single column of four black squares on the diagonal.

## Block separable form

$$\begin{bmatrix} P_{AA} & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_B^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_B \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & P_{BB} & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & P_{CC} & u_C \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_C^T & P_{DD} \end{bmatrix}$$



$$P_{i,j} = u_i S_{i,j} v_j^T$$

$$P^{-1} = E (S + \hat{D})^{-1} F^T + G$$

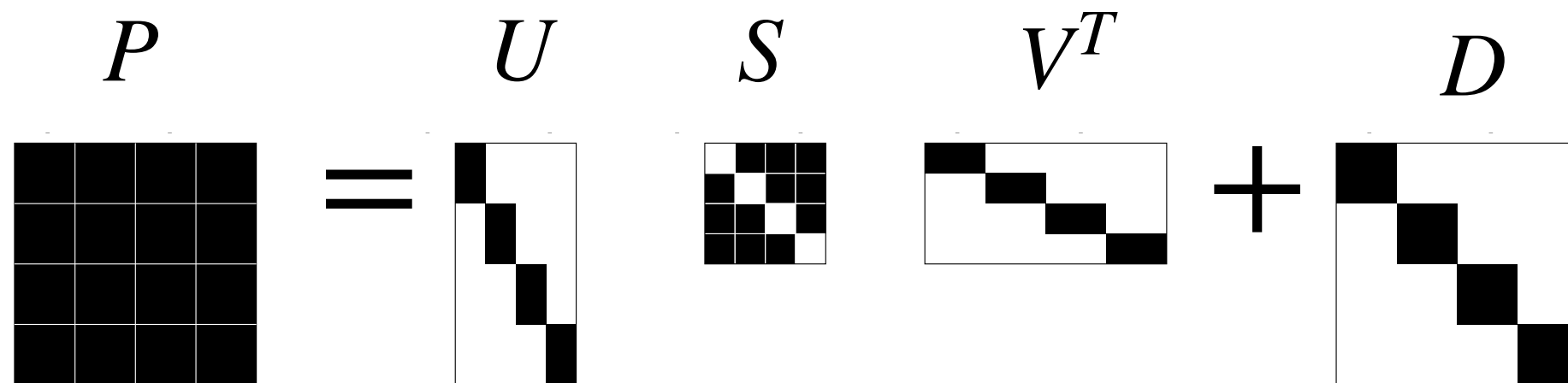
$$\hat{D} = (V^T D^{-1} U)^{-1}, \quad E = D^{-1} U \hat{D}, \quad F = (\hat{D} V^T D^{-1})^T, \quad G = D^{-1} - D^{-1} U \hat{D} V^T D^{-1}$$

Recall inverse with two intervals:

$$P^{-1} = \begin{bmatrix} P_{AA}^{-1} & 0 \\ 0 & P_{BB}^{-1} \end{bmatrix} - \begin{bmatrix} P_{AA}^{-1} u_A & 0 \\ 0 & P_{BB}^{-1} u_B \end{bmatrix} \left( I_2 + \begin{bmatrix} v_A^T P_{AA}^{-1} u_A & 0 \\ 0 & v_B^T P_{BB}^{-1} u_B \end{bmatrix} \right)^{-1} \begin{bmatrix} v_A^T P_{AA}^{-1} & 0 \\ 0 & v_B^T P_{BB}^{-1} \end{bmatrix}$$

## Sparse matrix embedding of P

$$P\sigma = f$$

$$P = U S V^T + D$$


$$\phi = V^T \sigma, \quad \psi = S \phi$$

$$\begin{bmatrix} V^T & 0 \\ S & -I \\ -I & S \end{bmatrix} \begin{bmatrix} \sigma \\ \psi \\ \phi \end{bmatrix} = \begin{bmatrix} f \\ 0 \\ 0 \end{bmatrix}$$

Matvec:  $O(N^2/4)$

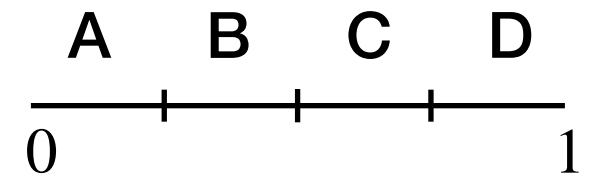
Inverse:  $O(N^3/32)$

**Not fast enough!**

S admits a similar factorization

## Structure of S

$$\begin{bmatrix} P_{AA} & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_B^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_B \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & P_{BB} & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_C^T & u_B \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_C \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & P_{CC} & u_C \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} v_D^T \\ u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_A^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_B^T & u_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v_C^T & P_{DD} \end{bmatrix}$$



$$P_{i,j} = u_i S_{i,j} v_j^T$$

$$S = \begin{bmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} (1 \ 0 \ 1 \ 0) \\ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} (0 \ 1 \ 0 \ 1) & \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \end{bmatrix}$$

## Heirarchical block separable form for P

$$P = U^{(\ell)} \left( U^{(\ell-1)} \left( \dots U^{(1)} D^{(0)} (V^{(1)})^T + D^{(1)} \right) \dots (V^{(\ell-1)})^T + D^{(\ell-1)} \right) (V^{(\ell)})^T + D^{(\ell)}$$

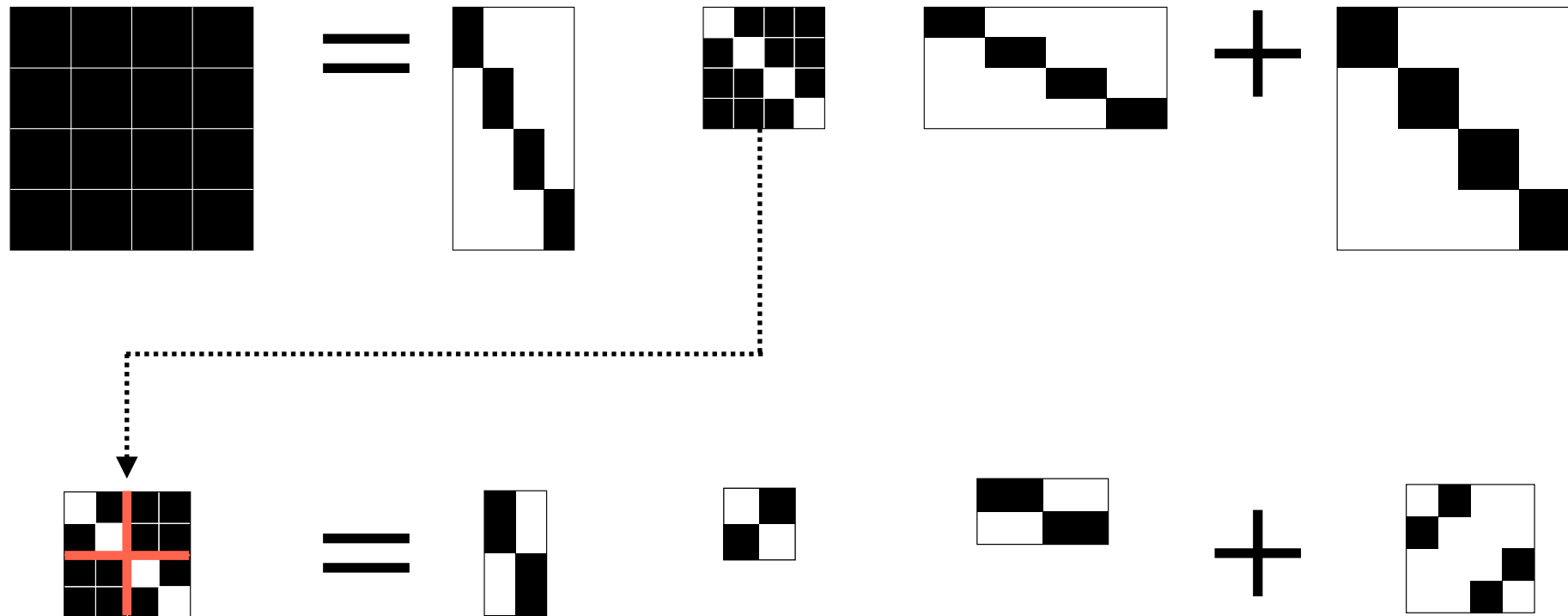
$P$

$U$

$S$

$V^T$

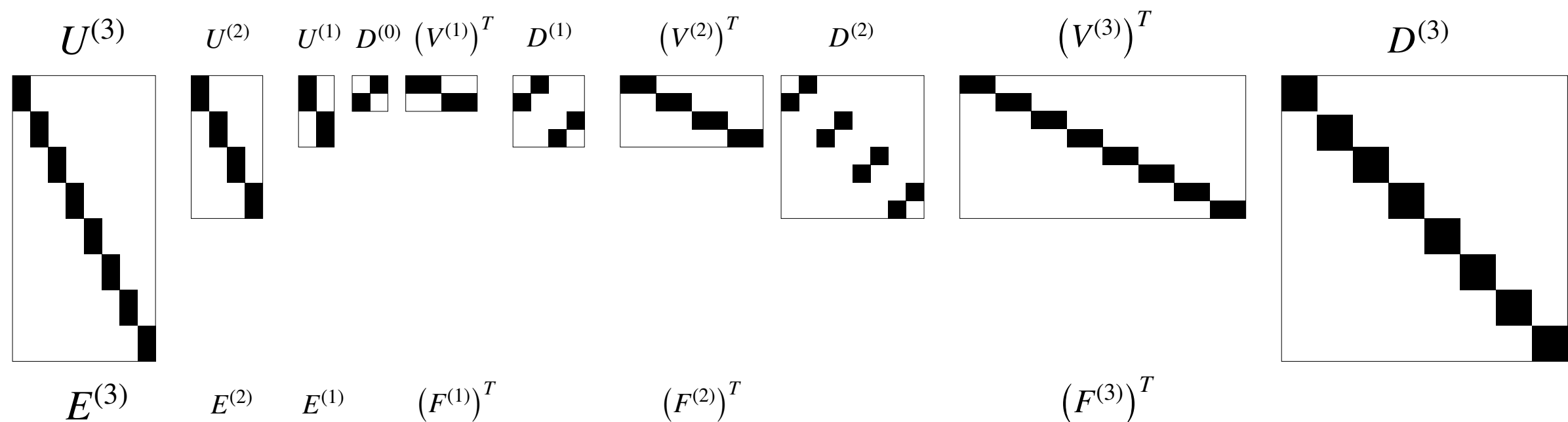
$D$



## Hierarchical block separable (HBS) form of P

$$P = U^{(3)} \left( U^{(2)} \left( U^{(1)} D^{(0)} (V^{(1)})^T + D^{(1)} \right) (V^{(2)})^T + D^{(2)} \right) (V^{(3)})^T + D^{(3)}$$

$$P^{-1} = E^{(3)} \left( E^{(2)} \left( E^{(1)} G^{(0)} (F^{(1)})^T + G^{(1)} \right) (F^{(2)})^T + G^{(2)} \right) (F^{(3)})^T + G^{(3)}$$

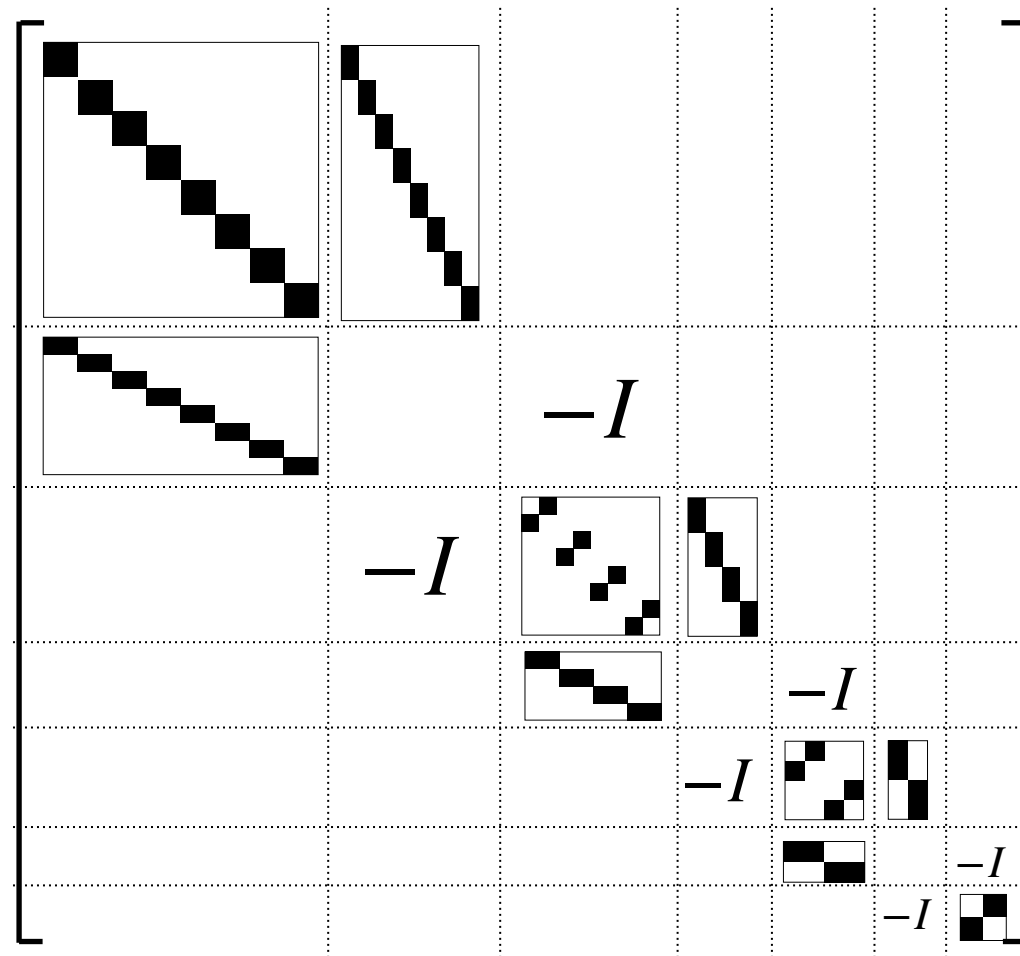
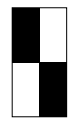
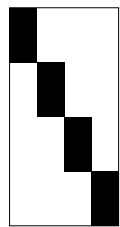
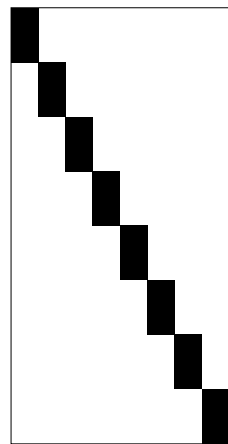


All matrices except  $G^{(0)}$  are block diagonal

# Sparse embedding of HBS form of P

$$P = U^{(3)} \left( U^{(2)} \left( U^{(1)} D^{(0)} (V^{(1)})^T + D^{(1)} \right) (V^{(2)})^T + D^{(2)} \right) (V^{(3)})^T + D^{(3)}$$

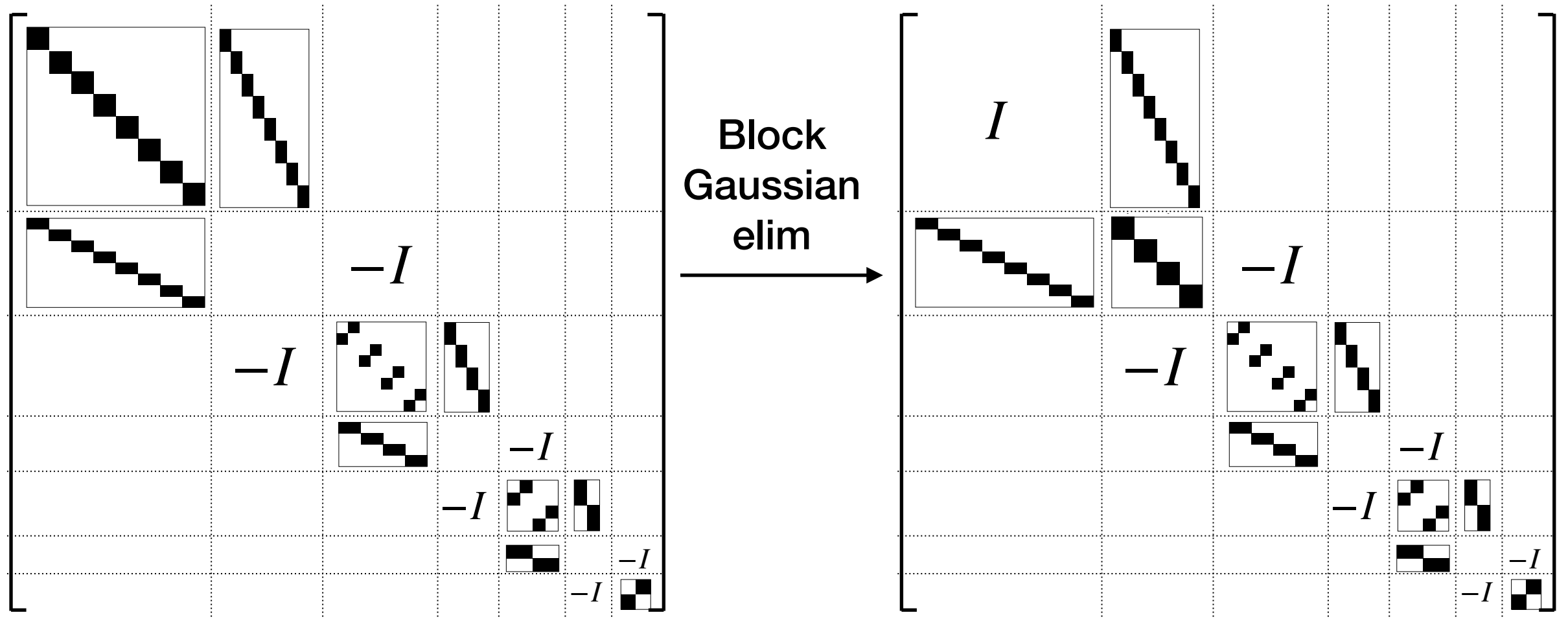
$U^{(3)}$      $U^{(2)}$      $U^{(1)}$      $D^{(0)}$      $(V^{(1)})^T$      $D^{(1)}$      $(V^{(2)})^T$      $D^{(2)}$      $(V^{(3)})^T$      $D^{(3)}$



Matvec:  $O(N)$

## Sparse embedding of HBS form of P - inversion

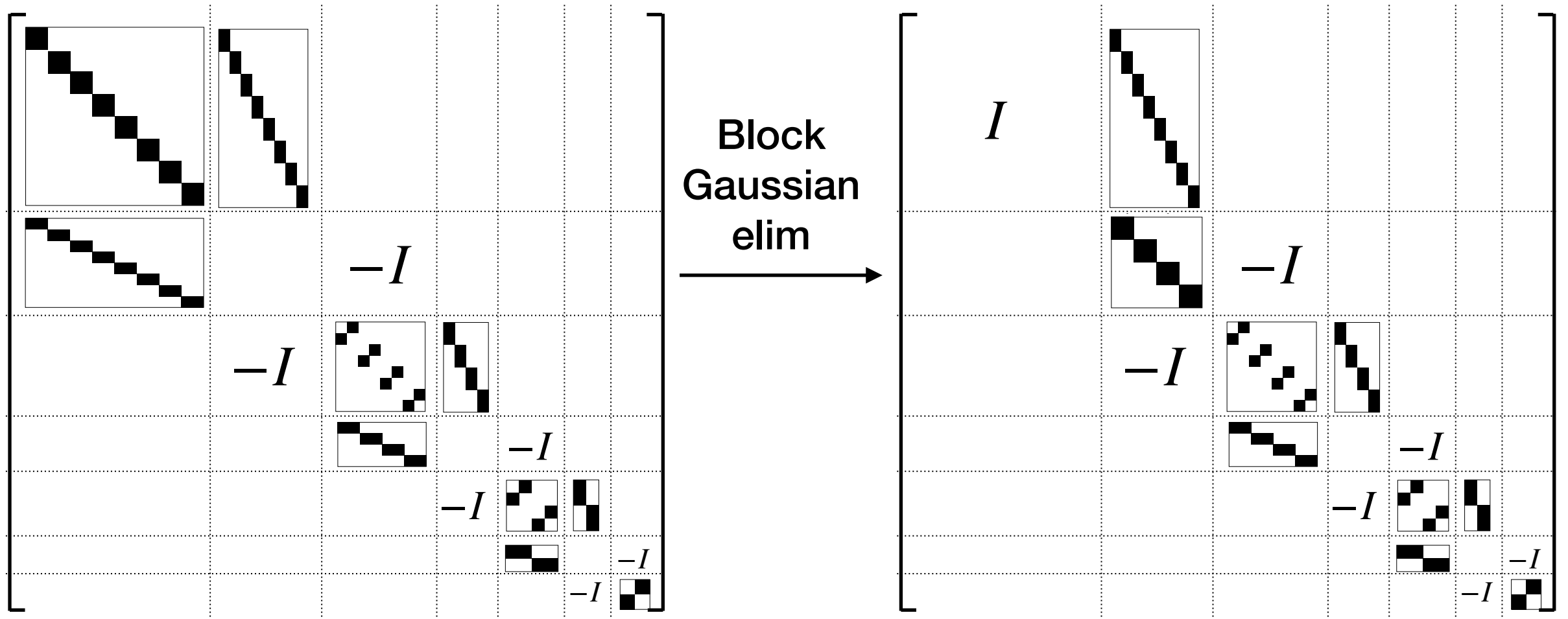
$$P^{-1} = E^{(3)} \left( E^{(2)} \left( E^{(1)} G^{(0)} (F^{(1)})^T + G^{(1)} \right) (F^{(2)})^T + G^{(2)} \right) (F^{(3)})^T + G^{(3)}$$



Inversion:  $O(N)$

## Sparse embedding of HBS form of P - inversion

$$P^{-1} = E^{(3)} \left( E^{(2)} \left( E^{(1)} G^{(0)} (F^{(1)})^T + G^{(1)} \right) (F^{(2)})^T + G^{(2)} \right) (F^{(3)})^T + G^{(3)}$$



Inversion:  $O(N)$

# Higher dimensions?

Given  $f(x)$ , find  $u(x)$  which satisfies

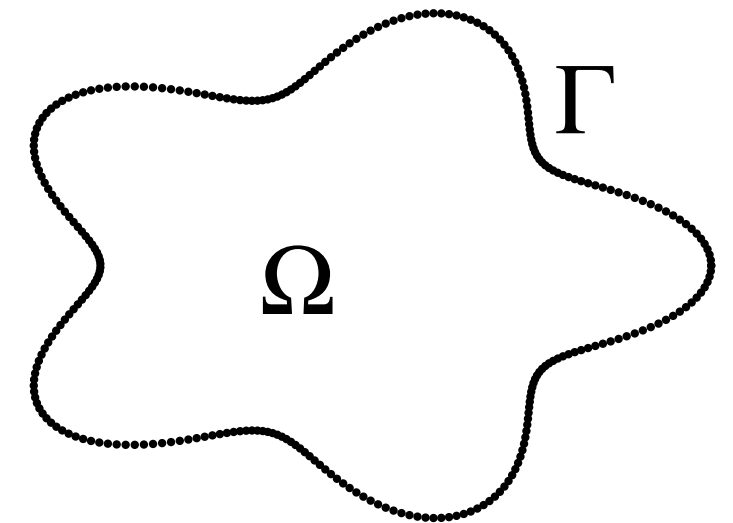
$$\Delta u(x) = 0 \quad x \in \Omega.$$

$$u(x) = f(x), \quad x \in \Gamma.$$

$G(x, y)$  : Green's function for

$$\Delta u(x) = \delta_y$$

$$G(x, y) = -\frac{1}{2\pi} \log |x - y|$$



**Integral formulation:**  $u(x) = \int_{\Gamma} \frac{\partial}{\partial \nu} G(x, y) \sigma(y) \quad x \in \Omega, \quad \sigma \text{ unknown density}$

- PDE ✓

$$\int_{\Gamma} \frac{\partial}{\partial \nu} G(x, y) \sigma(y) dS_y = \operatorname{Re} \int_{\Gamma} \frac{\sigma(\xi)}{z - \xi} d\xi \quad z = x_1 + ix_2, \quad \xi = y_1 + iy_2$$

- Boundary conditions yields following integral equation

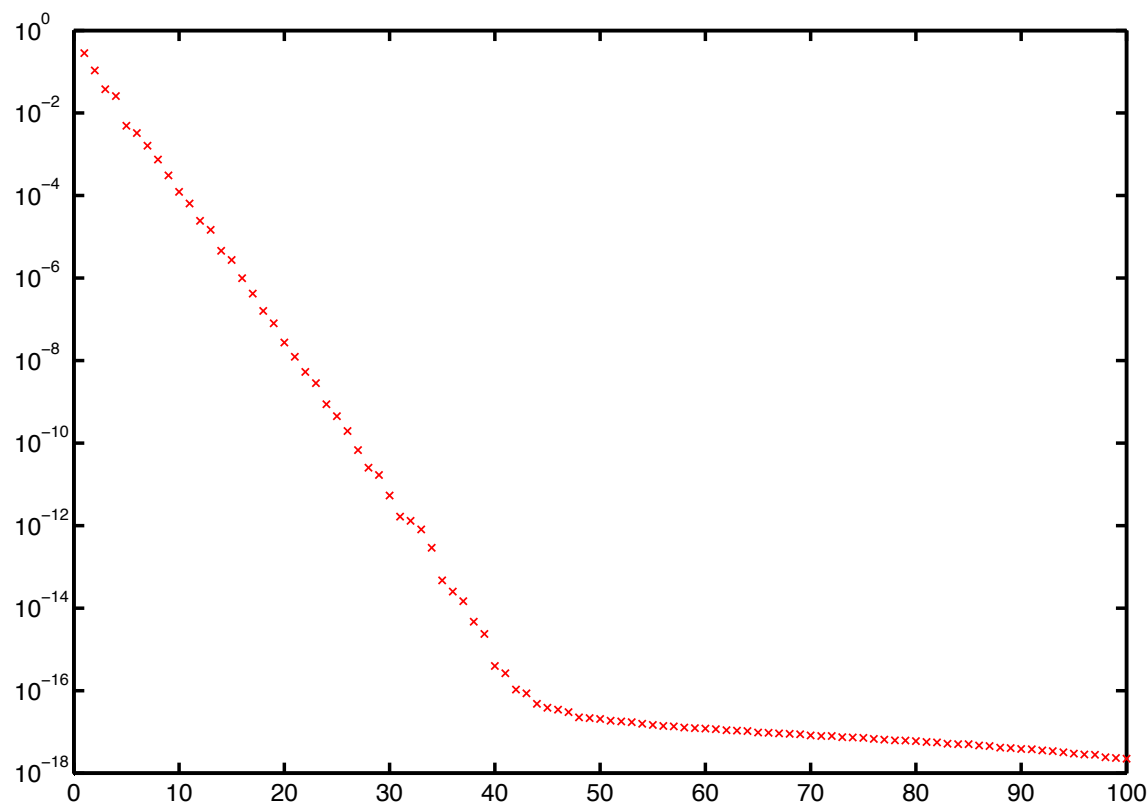
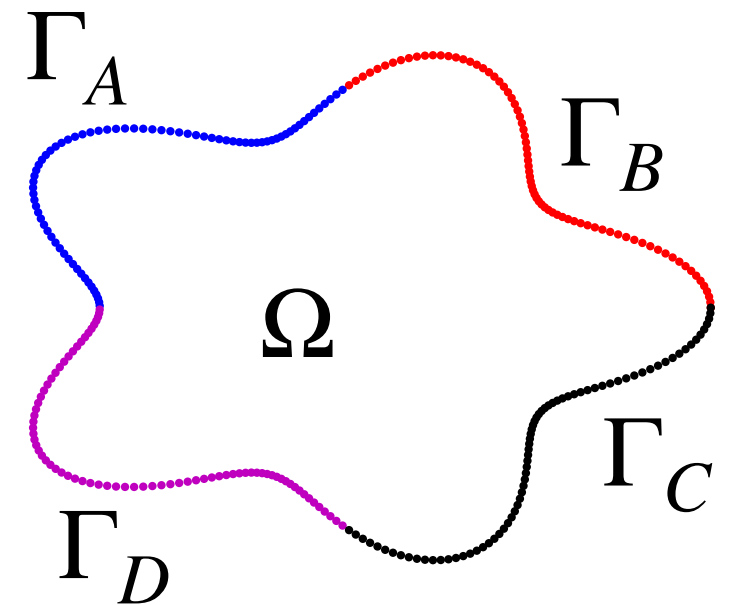
$$\sigma(x) + 2 \int_{\Gamma} \frac{\partial}{\partial \nu} G(x, y) \sigma(y) dS_y = 2f(x)$$

$$P\sigma = f$$

**P is also HBS compressible!**

# Off-diagonal blocks

$$\begin{bmatrix} P_{AA} & P_{AB} & P_{AC} & P_{AD} \\ \color{red}P_{BA} & P_{BB} & \color{red}P_{BC} & \color{red}P_{BD} \\ P_{CA} & P_{CB} & P_{CC} & P_{CD} \\ P_{DA} & P_{DB} & P_{DC} & P_{DD} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \\ \sigma_C \\ \sigma_D \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \\ \tilde{f}_D \end{bmatrix}$$



Singular values of matrix :  $\begin{bmatrix} P_{BA} & 0 & P_{BC} & P_{BD} \end{bmatrix}$

$$P_{i,j} = u_i S_{i,j} v_j^T$$

$u_i, v_j$ , now rank-k matrices

**How to compute  $u_i, v_j$ ?**

Option 1: Use analytical FMM expansions — matrix no longer HBS then, but  $\mathcal{H}^2$

Option 2: Use numerical compression techniques, like SVD, ID

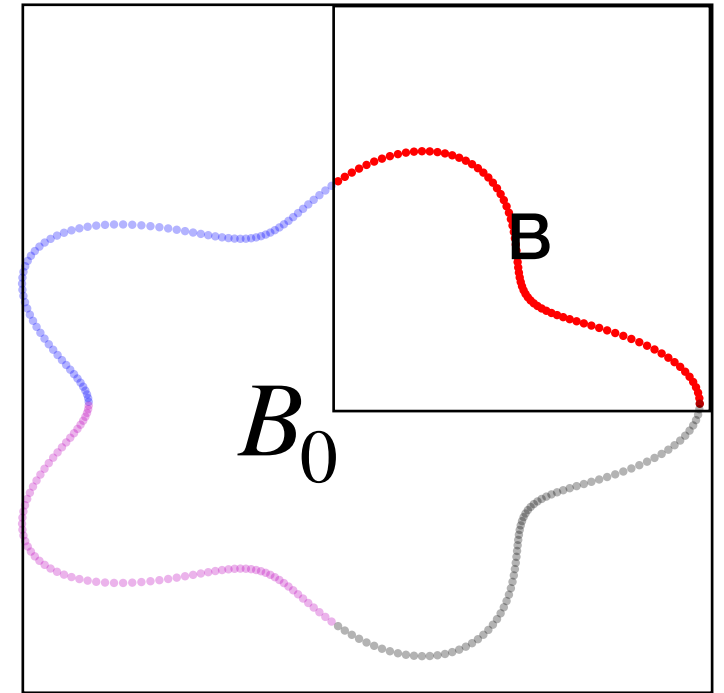
# Low-rank approximations - Functional SVDs

Suppose  $P\sigma = \sum_j K(x_i, y_j)\sigma_j$ ,  $x_i \in B$   $y_j \in B_0 \setminus B$

$Tf = \int_B K(x, y)f(y)dy$ ,  $T : \mathbb{L}^2(B_0 \setminus B) \rightarrow \mathbb{L}^2(B)$  with

$$\int_{B_0 \setminus B} \int_B |K(x, y)|^2 dx dy < \infty$$

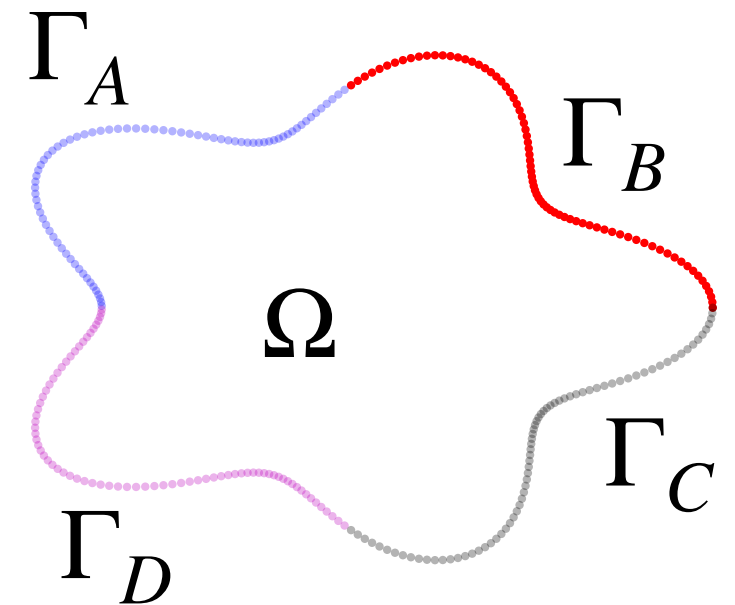
Then,  $K(x, y) = \sum_{i=1}^p u_i(x)s_i v_i(y) + O(\varepsilon)$ .



- Computing the functional SVD can be numerically intensive, particularly beyond  $d=2,3$
- Costs can be amortized for translationally invariant kernels  $K(|x - y|)$  and/or homogeneous kernels  $K(\lambda x, \lambda y) = \lambda^r K(x, y)$
- Computational savings if kernel satisfies Green's identities (Proxy surfaces)
- FMM-like translation operators through SVDs for recompressing S

# Low rank approximations - Randomized algorithms

$$\begin{bmatrix} P_{AA} & P_{AB} & P_{AC} & P_{AD} \\ \color{red}{P_{BA}} & P_{BB} & \color{red}{P_{BC}} & \color{red}{P_{BD}} \\ P_{CA} & P_{CB} & P_{CC} & P_{CD} \\ P_{DA} & P_{DB} & P_{DC} & P_{DD} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \\ \sigma_C \\ \sigma_D \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \\ \tilde{f}_D \end{bmatrix}$$



$$\begin{bmatrix} P_{AB} & 0 & P_{CB} & P_{DB} \end{bmatrix} = U_B \begin{bmatrix} \tilde{V}_{BA}^T & 0 & \tilde{V}_{BC}^T & \tilde{V}_{BD}^T \end{bmatrix}$$

$$\begin{matrix} p \times N & k \times N & k \times p \end{matrix}$$

**Randomized algorithms:**

$W = \mathbb{R}^{N \times (k+r)}$ , random Gaussian matrix, FFT matrix

$Y = P_B W$        $Y \in \mathbb{R}^{p \times (k+r)}$       Sample range of matrix

$Y = QR$       Orthogonalize sampled range

$T = Q^* P_B$        $T \in \mathbb{R}^{(k+r) \times p}$       Change of basis

$T = \hat{U} S V^T$       SVD of reduced matrix

$P_B \approx Q \hat{U} S V^T$

# Randomized algorithms - error analysis and performance

$$\begin{array}{c} P_B \\ \left[ \begin{array}{cccc} P_{AB} & 0 & P_{CB} & P_{DB} \end{array} \right] = U_B \left[ \begin{array}{cccc} \tilde{V}_{BA}^T & 0 & \tilde{V}_{BC}^T & \tilde{V}_{BD}^T \end{array} \right] \\ \begin{array}{ccc} p \times N & k \times N & k \times p \end{array} \end{array}$$

**Randomized algorithms:**

$W = \mathbb{R}^{N \times (k+r)}$ , random Gaussian matrix, FFT matrix

$Y = P_B W$        $Y \in \mathbb{R}^{p \times (k+r)}$       Sample range of matrix       $O(N \cdot (k+r) \cdot p)$

$Y = QR$       Orthogonalize sampled range       $O(p \cdot (k+r)^2)$

$T = Q^* P_B$        $T \in \mathbb{R}^{(k+r) \times N}$       Change of basis       $O(N \cdot (k+r) \cdot p)$

$T = \hat{U} S V^T$       SVD of reduced matrix       $O((k+r)^2 \cdot N)$

$P_B \approx Q \hat{U} S V^T$

$$\|P_B - Q \hat{U} S V^T\| = \|P_B - QT\| = \|P_B - QQ^* P_B\|$$

$$\|P_B - QQ^* P_B\| \leq \left(1 + C\sqrt{N}\right) s_{k+1} \quad \text{with probability } 1 - 6r^{-r}$$

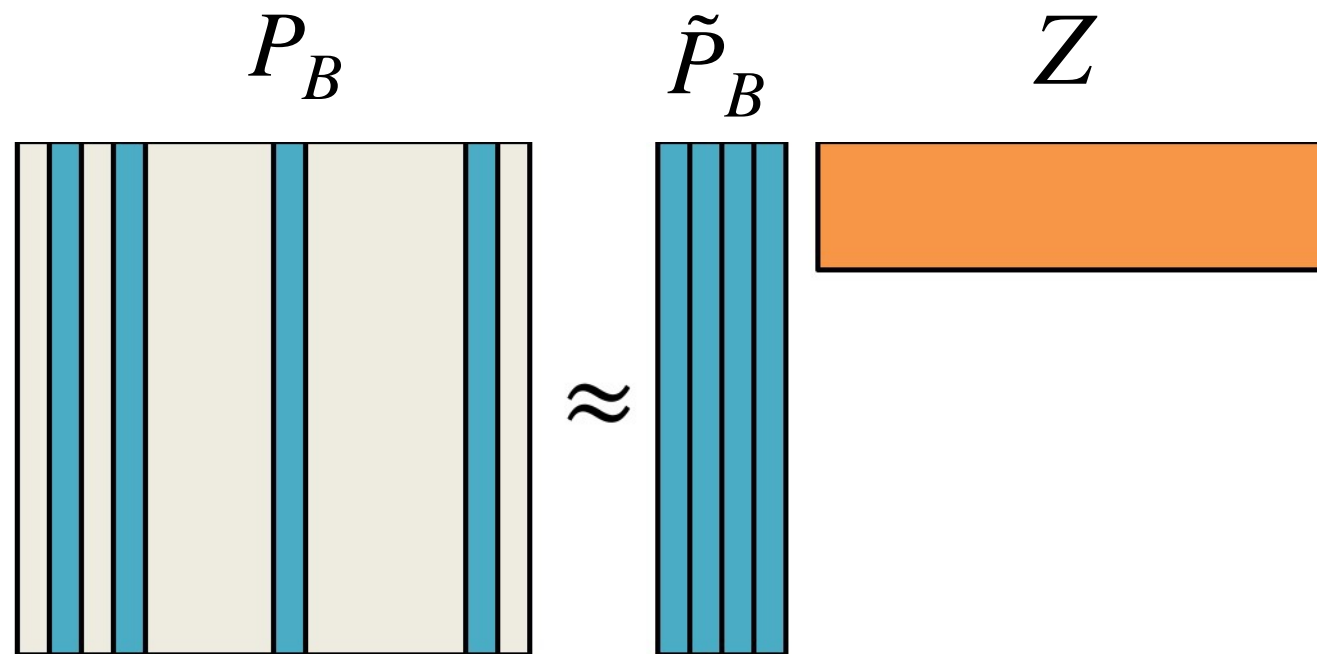
**Issues:**

Cost of compressing one block of rows:  $O(N \cdot (k+r) \cdot p)$

N such factorizations needed  $\implies$  cost of factorization:  $O(N^2)$

Lack of interpretability of S at next layer

# Interpolative Decomposition (ID)



$$\|P_B - \tilde{P}_B Z\| \leq (1 + \sqrt{k(n-k)})s_{k+1}$$

$$|Z_{i,j}| \leq 1$$

Combinatorial search, exponential cost

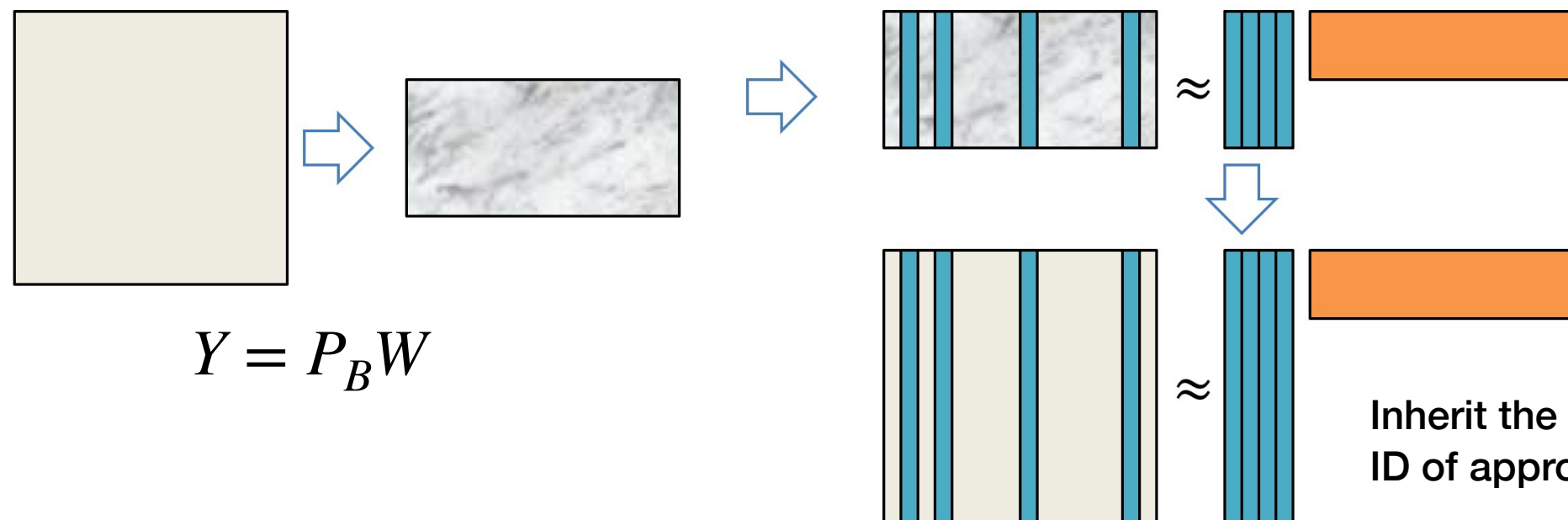
$$\|P_B - \tilde{P}_B Z\| \leq (1 + \sqrt{k(n-k)})s_{k+1}$$

$$|Z_{i,j}| \leq 2$$

$$O(N \cdot p^2 \log N)$$

In practice, rank revealing QR works fine

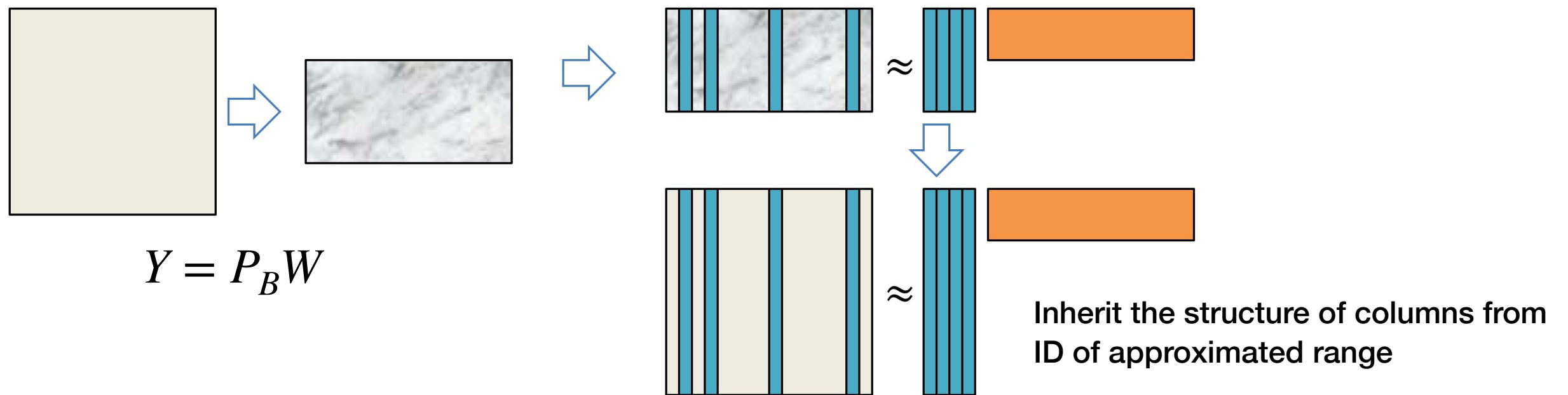
Low rank approximation that uses columns of input matrix



Inherit the structure of columns from ID of approximated range

Randomized approach for computing ID

# Interpolative Decomposition (ID)



Randomized approach for computing ID

## Issues:

Cost of compressing one block of rows:  $O(N \cdot (k + r) \cdot p)$

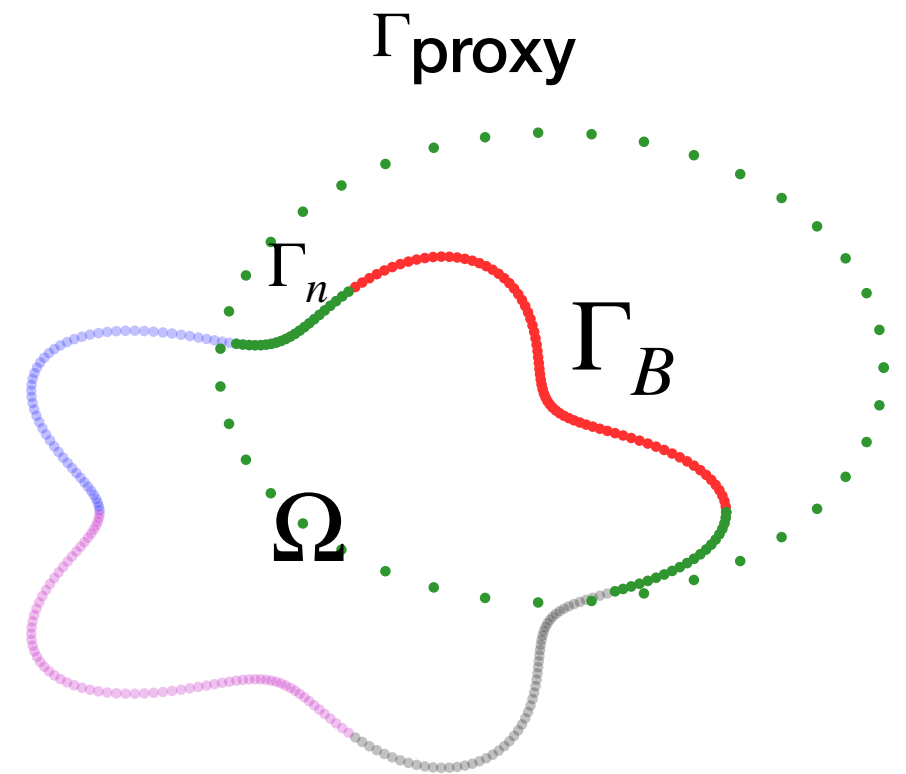
N such factorizations needed  $\implies$  cost of factorization:  $O(N^2)$

~~Lack of interpretability of S at next layer~~

Entries of S are sub-blocks of the original matrix

# Low rank approximations - Proxy surfaces

$$\begin{bmatrix} P_{AA} & P_{AB} & P_{AC} & P_{AD} \\ \color{red}{P_{BA}} & P_{BB} & \color{red}{P_{BC}} & \color{red}{P_{BD}} \\ P_{CA} & P_{CB} & P_{CC} & P_{CD} \\ P_{DA} & P_{DB} & P_{DC} & P_{DD} \end{bmatrix} \begin{bmatrix} \sigma_A \\ \sigma_B \\ \sigma_C \\ \sigma_D \end{bmatrix} = \begin{bmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \\ \tilde{f}_D \end{bmatrix}$$



Instead of compressing  $P_B$ , compress  $\begin{bmatrix} P_{B,\Gamma_n} & P_{B,\Gamma_{\text{proxy}}} \end{bmatrix}$   
 $p \times (n_{\text{proxy}} + n_{\text{near}})$

Works when matrix entries from Kernel satisfying Green's identity

General idea: identify smaller collection of columns which approximate bulk of matrix

## Issues:

~~Cost of compressing one block of rows:  $O(N \cdot (k + r) \cdot p)$~~

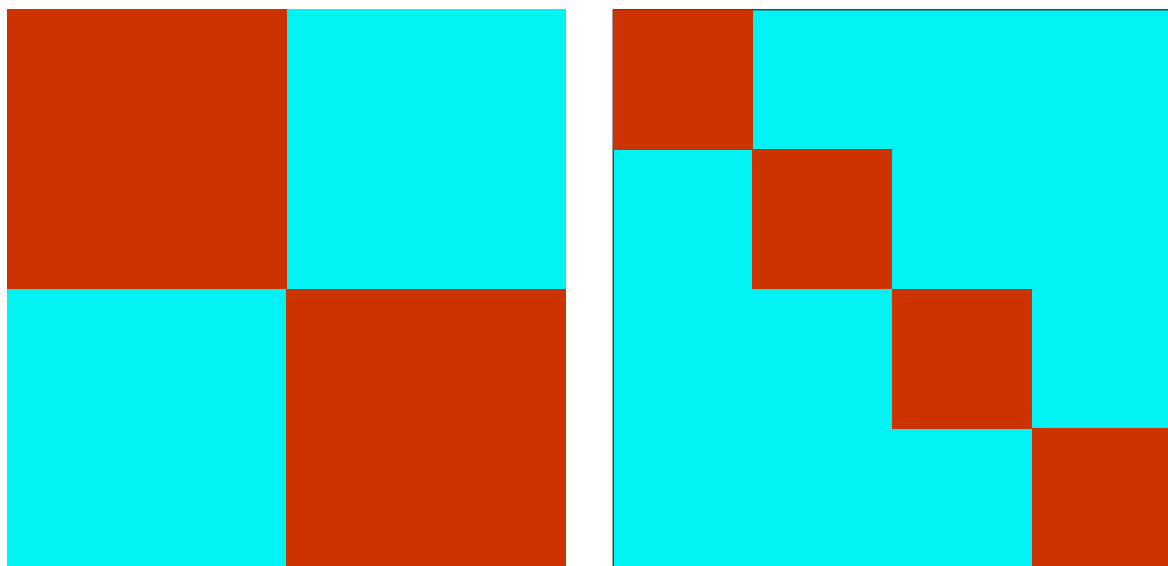
Cost of compressing one block of rows:  $O((n_{\text{proxy}} + n_{\text{near}}) \cdot (k + r) \cdot p)$

~~N such factorizations needed  $\Rightarrow$  cost of factorization:  $O(N^2)$~~

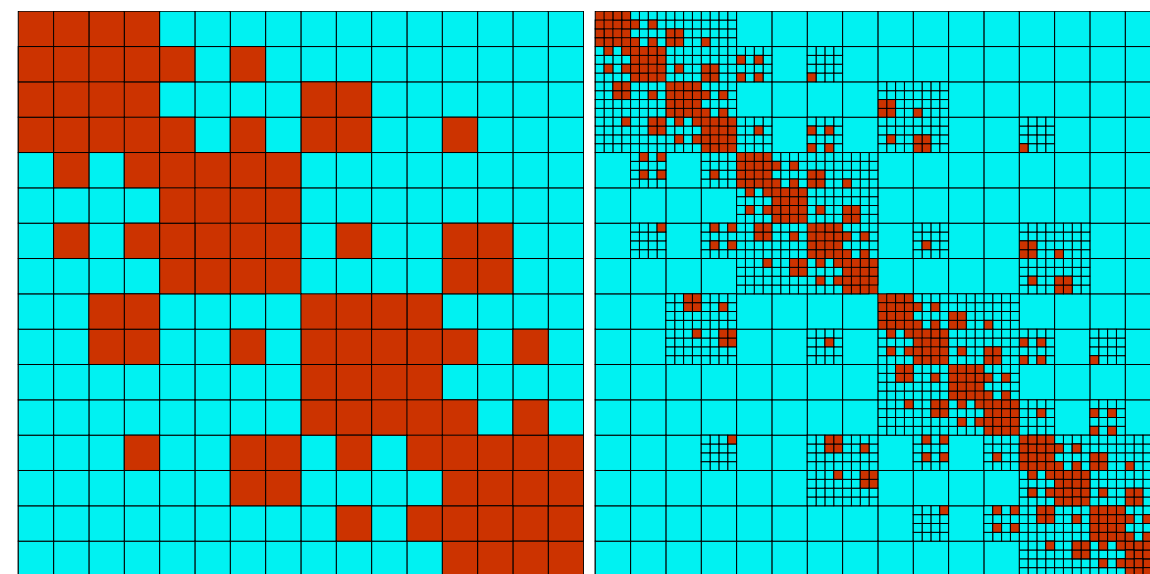
~~Lack of interpretability of S at next layer~~

Entries of S are sub-blocks of the original matrix

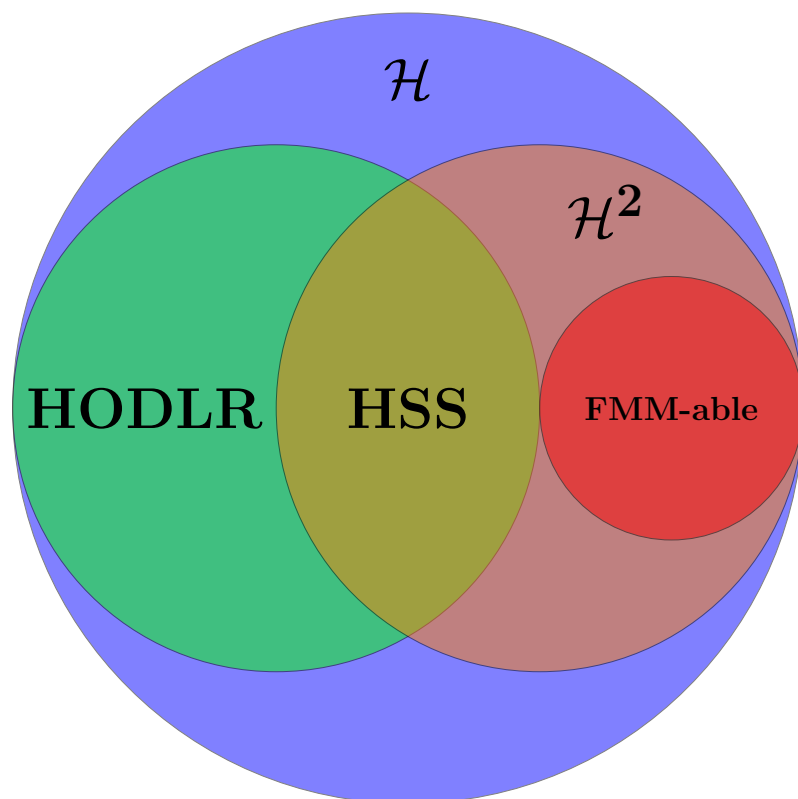
# The zoo of matrix factorizations



**HODLR/HSS matrices**



**FMM/ $\mathcal{H}^2$  matrices**



**Butterfly/FFT matrices**

Low-rank structure  
↓

|        | Nested basis → |                 |
|--------|----------------|-----------------|
|        | No             | Yes             |
| Strong | HODLR          | HSS             |
| Weak   | $\mathcal{H}$  | $\mathcal{H}^2$ |

# Hierarchical matrices in Neural Networks

A multiscale neural network based on hierarchical nested bases

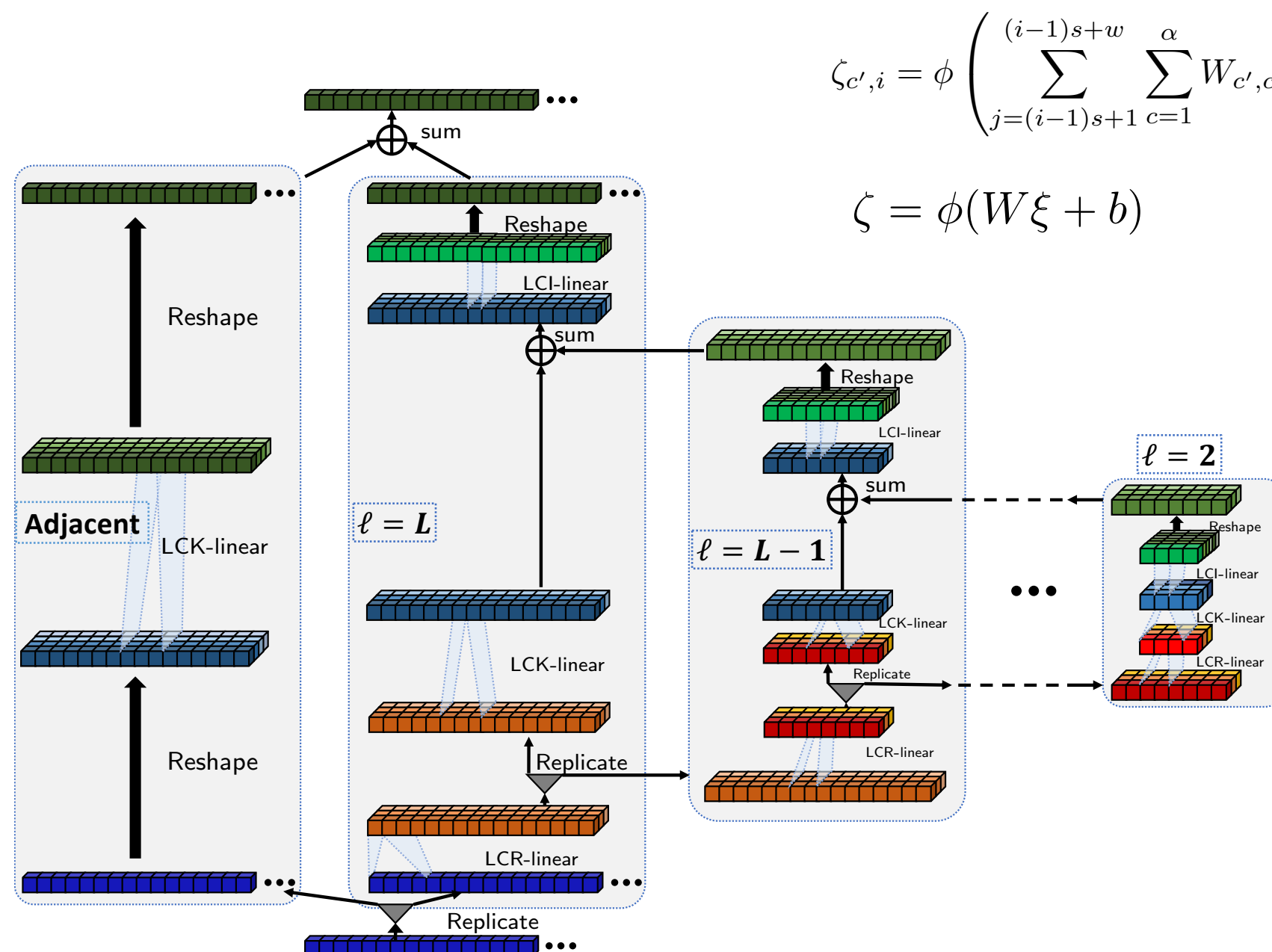
Yuwei Fan\*, Jordi Feliu-Fabà†, Lin Lin‡, Lexing Ying§, Leonardo Zepeda-Núñez¶

Using  $\mathcal{H}^2$  in layers of locally connected networks

A multiscale neural network based on hierarchical matrices

Yuwei Fan\*, Lin Lin‡, Lexing Ying‡, Leonardo Zepeda-Núñez§

Using  $\mathcal{H}$  in layers of locally connected networks

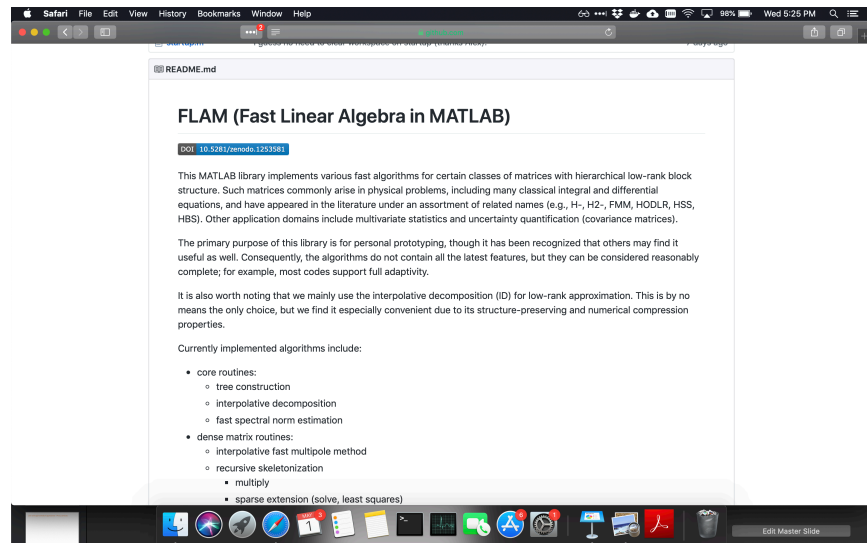


$$\zeta_{c',i} = \phi \left( \sum_{j=(i-1)s+1}^{(i-1)s+w} \sum_{c=1}^{\alpha} W_{c',c;i,j} \xi_{c,j} + b_{c',i} \right), \quad i = 1, \dots, N'_x, \quad c' = 1, \dots, \alpha'$$

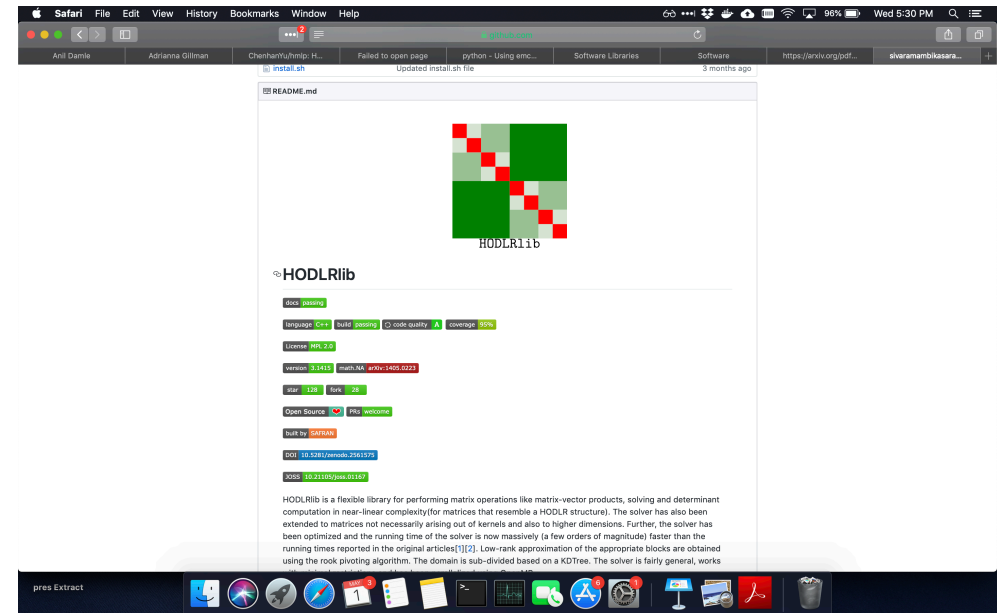
$$\zeta = \phi(W\xi + b)$$

Affords fewer number of parameters in Neural net representation and fast application of the forward network

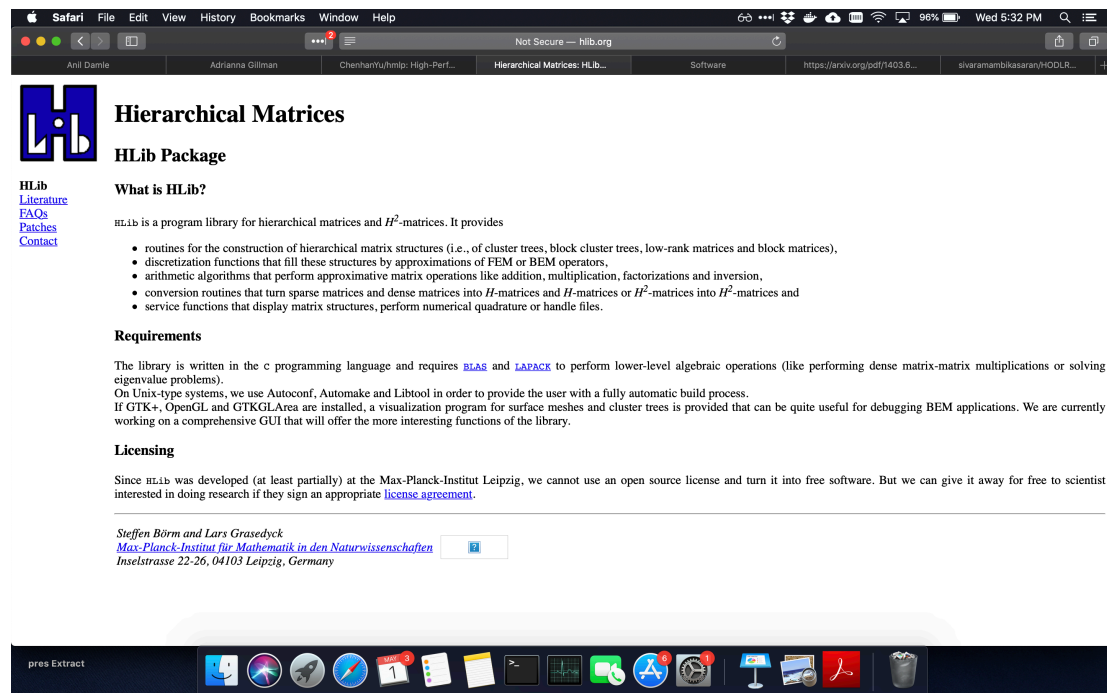
# Software



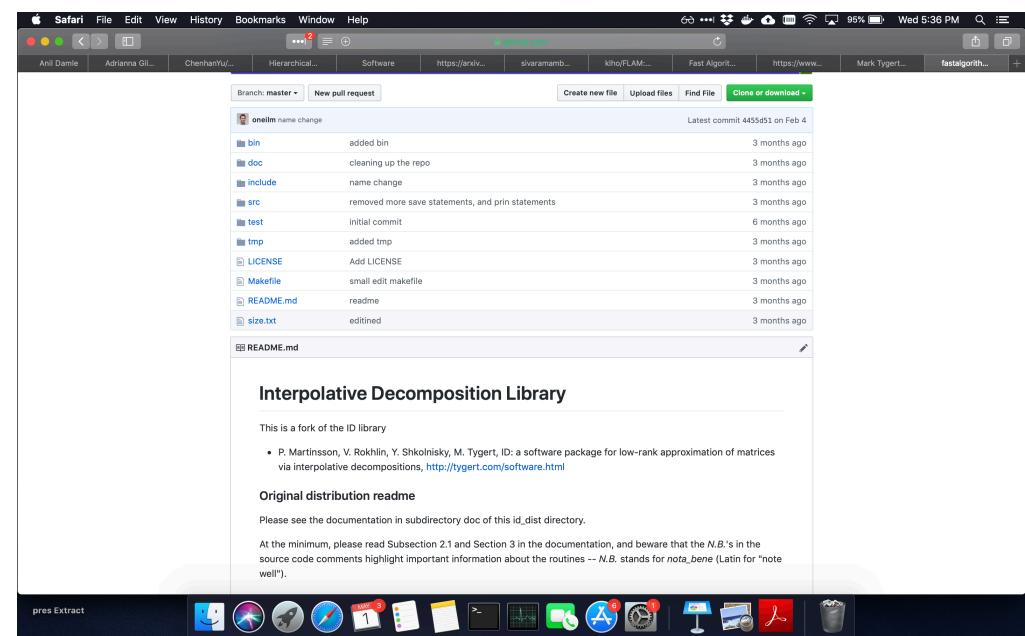
<https://github.com/klho/FLAM>



<https://github.com/sivaramambikasaran/HODLR>

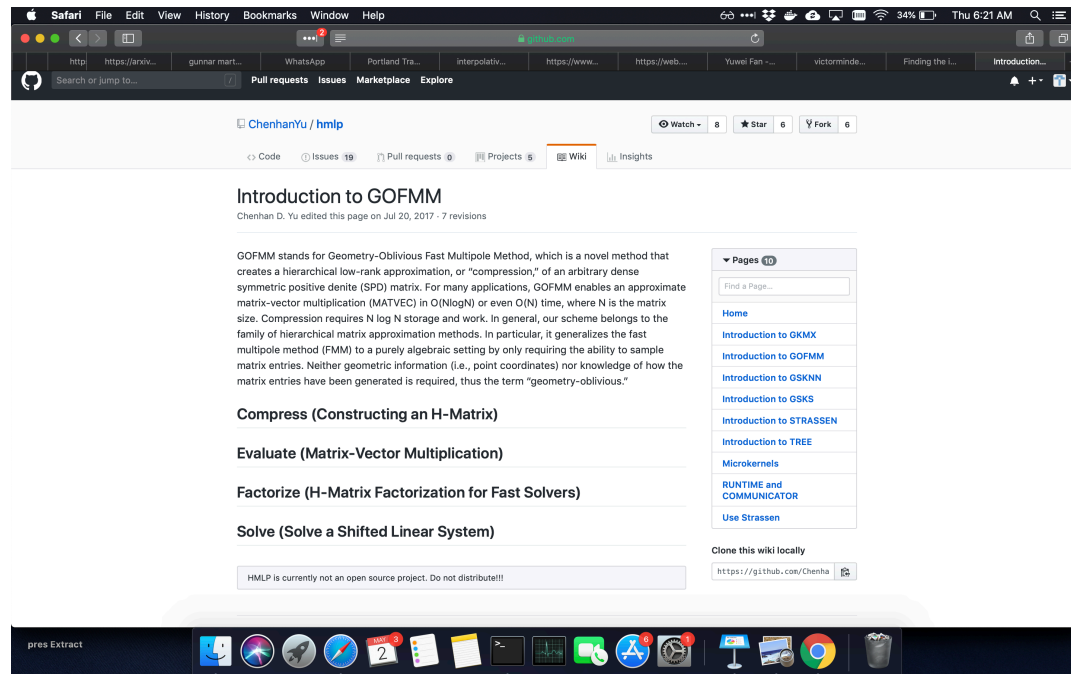


<http://www.hlib.org>

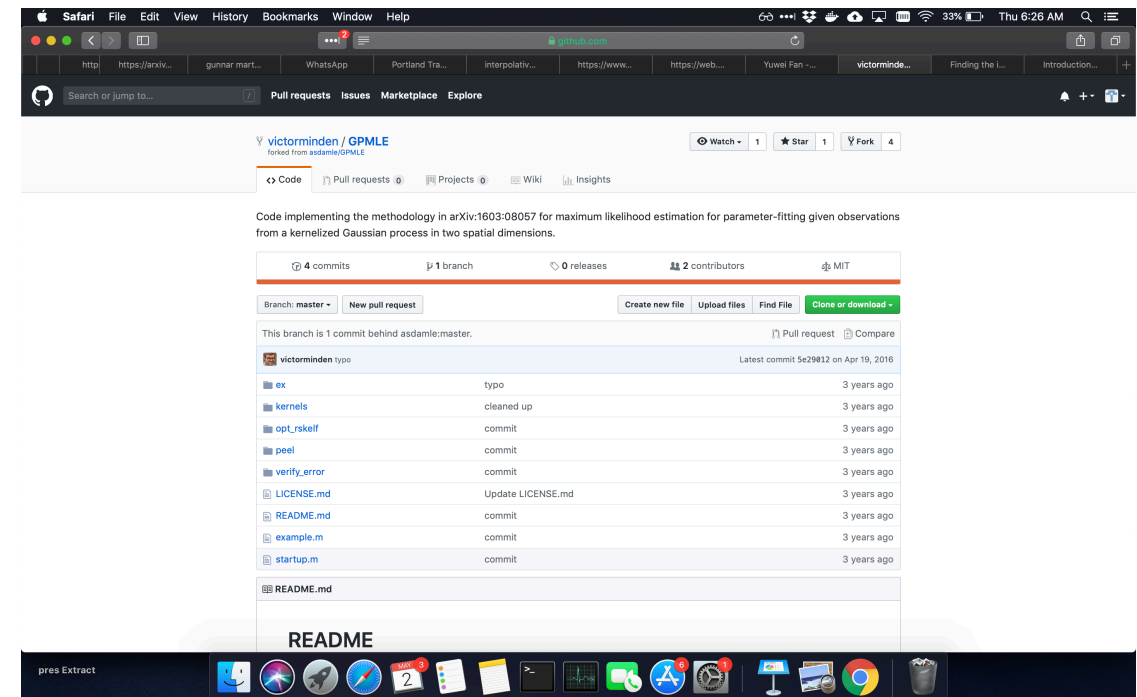


<https://github.com/fastalgorithms/libid>

# More resources



<https://github.com/ChenhanYu/hmlp/wiki/Introduction-to-GOFMM>



<https://github.com/victorminden/GPMLE>

- Video lectures by Gunnar - [https://www.youtube.com/playlist?list=PLPDZ9rcIfxyOrlpcu\\_D1PRcyK-o2iofwW](https://www.youtube.com/playlist?list=PLPDZ9rcIfxyOrlpcu_D1PRcyK-o2iofwW)
- Excellent review article on randomized methods for low rank approximations - Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions: <https://arxiv.org/pdf/0909.4061.pdf>
- Some of the illustrations courtesy: Sivaram Ambikasaran, Per-Gunnar Martinsson, Ken Ho, Leslie Greengard, Lexing Ying, Adrianna Gillman

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**Not an exhaustive list**